Answer Sheet

Write your answers on this sheet. Exam questions & instructions can be found on subsequent sheets.

Name:

Student ID number:

Answers to knowledge questions:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	F	F	F	F	T	F	F	T	F	F	T	F	F	T	F	T	T	T	F	T
B	T	T	F	F	F	T	T	F	T	F	F	T	F	F	T	T	F	T	T	F
C	F	T	T	F	T	F	T	F	F	T	F	F	T	F	T	T	T	T	F	F
D	F	F	T	T	F	T	T	F	F	T	F	F	F	T	F	T	F	F	T	F

Answers to hands-on questions:

	21	22	23	24	25	26	27	28	29	30
A	F	F	T	F	T	F	F	T	F	F
B	T	F	F	F	T	T	T	F	F	F
C	F	F	F	T	T	T	T	F	T	T
D	T	T	F	T	T	F	F	F	F	F

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valM	`feedfacebeef0bad`
R[rsp]	`6306`
R[rA]	`feedfacebeef0bad`
PC	`202420`

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VPN	16
TLB index	2
TLB tag	5
TLB hit	T
Page fault	T
PPN	NA

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A	25
B	8

34

	SU	ST	PU	PT
SU	rw	r	r	r
ST	w	rw	-	r

Answers to analytical questions:

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Programs that have a stride-1 memory access pattern follow the principle of locality, and thus make efficient use of cache memory. This improves performance.

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One solution (a justification):
To implement a mutex, you need to serialize access to a state. The *fetch-and-add* instruction is an atomic RMW (read-modify-write) to memory, that can be utilized to implement lock acquisition and release. Atomicity ensures that threads using this instruction to modify memory do so serially.

Another solution (an implementation):
To acquire a lock, *fetch-and-add* 1 to a “ticket number” word (thus incrementing it), and spin until a “turn” word equals it. Release a lock by incrementing the “turn” word.

End of answer sheet.